

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Analyzed complex datasets from diverse global sources, identifying key trends and patterns to inform strategic business decisions and AI initiatives.
- Developed and deployed predictive models using **Python** and **scikit-learn**, achieving an average **0.88 AUC** on held-out data for forecasting pharmaceutical demand.
- Collaborated with cross-functional teams, including business analysts and IT stakeholders, to gather requirements and translate them into actionable **data science** and **AI solutions**.
- Contributed to the design and implementation of **data pipelines** on **AWS**, processing large volumes of data for analytics and model training.
- Assisted in the evaluation and integration of emerging **AI technologies**, including prompt engineering techniques for **generative AI use cases**.
- Presented complex analytical findings and model performance metrics to both technical and non-technical stakeholders, ensuring clear understanding and buy-in.

SKILLS

Programming Languages – Python, R, SQL, Java

Data Science & ML – Machine Learning, AI, Predictive Modelling, NLP, Generative AI, Prompt Engineering, scikit-learn, TensorFlow, PyTorch, Data Governance, Model Governance

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Visualisation, Statistical Analysis, Exploratory Data Analysis

Cloud Platforms – AWS, Azure, GCP

Databases – PostgreSQL, MySQL, Oracle, SQL

Tools & Methodologies – Git, Jenkins, Jira, Agile, Scrum, Docker

Domain Knowledge – Pharmaceutical Industry, Animal Health

PROJECTS

Pharmaceutical Demand Forecasting Model

- Engineered a **Python**-based data pipeline to ingest historical sales and market data from disparate sources, cleaning and transforming it into a format suitable for modeling.
- Trained and evaluated a **LightGBM regression model** on ~1.5M transaction records to predict regional product demand, achieving an average **RMSE of 42.5 units**.
- Developed interactive **Power BI dashboards** to visualize forecast accuracy, key drivers, and confidence intervals for the sales and operations planning teams.

AI-Driven Inventory Optimization Analysis

- Collaborated with the supply chain team to define key performance indicators and extract relevant data from enterprise systems using **SQL**.
- Explored the application of **generative AI** techniques to identify potential patterns in inventory management data, generating hypotheses for further investigation.
- Supported the development of a proof-of-concept dashboard in **Power BI** to present initial findings on inventory discrepancies and potential optimization opportunities.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)